

## **Performance Evaluation of a Continuous Line Manufacturing System using Overall Equipment Effectiveness and Line Balancing**

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### **ABSTRACT**

*This project was undertaken to evaluate the performance of a continuous line manufacturing system using overall equipment effectiveness (OEE) and line balancing in a rice packaging company. The process adopted was to analyze data from the equipment in the production line, evaluate them using overall equipment effectiveness and improve the continuous line manufacturing system of a rice packaging company using the results as a source of the decisions to increase productivity. Also, line balancing was integrated to reduce the idling time of the equipment in workstations. To achieve this project, the following objectives were concluded, the evaluation of process parameters on availability, performance and quality and the total product of the process parameters gave the performance of the continuous line manufacturing system in a rice packaging company. The overall equipment effectiveness results obtained after collecting data for a shift length time is 14.69697% while the average availability, performance, quality are 92.35697%, 16.42210%, and 97% respectively. The significance of the performance evaluation of a rice packaging company using overall equipment effectiveness is to measure productivity of the equipment and improve efficiency. Line balancing was introduced to the rice packaging production line by creating workstations to reduce idling time, reduce waste and enhance workflow in the production line by 76.39%. In conclusion, the production line was improved by 61.69303 % after line balancing from 14.69697% to 76.39%.*

**Keywords:** Metrics, takt time, availability, performance, quality, downtime

### **INTRODUCTION**

Today, due to adoption of Continuous Line Manufacturing System of production and its benefits especially in mass production; This research was considered due to sudden failure of plant equipment and to measure the effectiveness and productivity of a rice packaging equipment to avoid waste in production.

Finding Centrifugal Pump Effectiveness Using OEE Measures as a Foundation for Optimal Process Stability was the subject of a study. The measurement of centrifugal oil pump effectiveness using OEE was

necessary to maintain the petroleum industry's competitiveness on a global scale. For increased efficiency and profitability, the OEE metric data is used to forecast the pump efficacy over a three-year period. [1].

A study was carried out to explain the Effect of Machine Efficiency in Production Line Balancing. They defined Production Line Balancing (PLB) as a technique of assigning operations to workstations in such a way that the assignment minimises the idle time

between workstations, increasing the line efficiency to 81.7% [2].

An investigation was conducted to elucidate the utilization of a novel measure, the Overall Equipment Effectiveness of a Manufacturing Line, as a cohesive method for assessing a production line's performance. [3].

### **GENERAL METHODOLOGY**

The material used to capture analogue data from the equipment in the continuous line manufacturing system of Rice packaging plant is the Stopwatch. While overall equipment effectiveness and Line Balancing are the metrics used to measure the equipment effectiveness.

The facility of the case study is a Rice packaging company in Uke, Karu LGA, Nasarawa State which processes and packages rice in a commercial quantity.

The following data were captured from the Rice packaging company system equipment: Shift Length Time (Sec or Minutes), Breaktime (Minutes), Downtime (Minutes), Ideal Cycle Time (50kg/minute), Total Count (kg), and Rejected Count (kg).

For performance valuation, the following steps are required;

#### **Calculating Planned Production Time**

The planned production time is the total time the production processes are scheduled to operate [4]. This can be obtained using the following equation.

$$\text{PPT} = \text{Shift Length Time} - \text{Break time} \quad (1)$$

$$\text{Performance} = (\text{Actual Output} / \text{Standard Output}) \times 100\% \quad (5)$$

#### **Calculating Quality Loss (Process Parameter 3):**

The final OEE factor that needs to be assessed is quality. It identifies the quality of the product (defect-free) [9]. This can be obtained using the following equation.

#### **Calculating Run Time**

Run time is the actual amount of time that production was running excluding downtime (planned stops and unplanned stops) [5]. This can be obtained using the following equation.

$$\text{Run Time} = \text{PPT} - \text{Downtime} \quad (2)$$

#### **Calculating the Good Count**

The Good Count is the unit of products that are defect-free from the Total Count [6]. This can be obtained using the following equation.

$$\text{Good Count} = \text{Total Count} - \text{Rejected Count} \quad (3)$$

#### **Calculating Availability Loss (Process Parameter 1):**

##### **Loss**

The first of the three overall equipment effectiveness aspects to be assessed is availability loss. It detects both planned and accidental stoppage when the process is not operating at its best [7]. This can be acquired by applying the subsequent formula.

$$\text{Availability} = (\text{Run Time} / \text{PPT}) \times 100\% \quad (4)$$

#### **Calculating Performance Loss (Process Parameter 2)**

The second of the three overall equipment effectiveness variables that need to be assessed is performance loss. It identifies when the process is running slower than its theoretical top speed (both idling & minor stops and reduced speed) [8]. This can be obtained using the following equation.

$$\text{Quality} = (\text{Good Count} / \text{Total Count}) \times 100\% \quad (6)$$

### Evaluating overall equipment effectiveness

Multiplying the three process parameter components mentioned above yields the evaluation performance [10]. The relationship that follows can be used to acquire this.

$$\text{OEE (\%)} = \text{Availability} \times \text{Performance} \times \text{Quality} \quad (7)$$

### Line Balancing Metric

The practice of allocating equal process time to each workstation and grouping production activities together in order to maximize equipment usage and reduce idle time is known as line balancing. [11].

Heuristic method of line balancing was adopted where the following procedural steps were involved;

- Identifying the work to be done (Rice processing and packaging)
- Breaking down the work into tasks
- Listing the various tasks involve (Parboiling, Drying, Destoning...etc)
- Sketching the precedence diagram (task schedule network diagram) and mark the task duration

Creating workstation to show maximum take time in each column [12].  
The result for line efficiency was obtained using the following equation.

$$\text{Line efficiency} = \frac{\text{sum of cycle time}}{\text{No. of workstations} \times \text{takt time}} \quad (8)$$

The result of balance delay time is obtained using the following equation [13].

$$B.D = \frac{\text{No. of workstations} \times \text{takt time} - \text{sum of cycle } T}{\text{No. of workstations} \times \text{takt time}} \quad (9)$$

## RESULTS AND DISCUSSION

The performance evaluation analysis in Table 1 expressed the results of the evaluation process from the project where the first column is the equipment in the production line of the rice packaging company, followed by key evaluation data, the process parameters availability, performance, quality and their OEE.

The overall equipment effectiveness is used to measure the productivity of the equipment of rice packaging line and from the Table 1 using Microsoft Excel spreadsheet to calculate, the availability, performance and quality of all the equipment were shown with their corresponding oee. The calculation first expressed the evaluation process

parameters using Equation (1) to determine planned production time, while the calculations in Equations (2) and (3) expressed run time and good count. The process parameters of OEE were evaluated in Equations (4, 5, and 6) where availability, performance and quality were determined respectively to measure oee on each equipment of the rice packaging line. Here the production line availability, performance, and quality average percentages are 92.69697, 16.42210 and 97 % respectively. This is showing when the process is running optimally, as fast as possible, and when there are no defects (good parts are being produced). The following are expressed in the Table 1.

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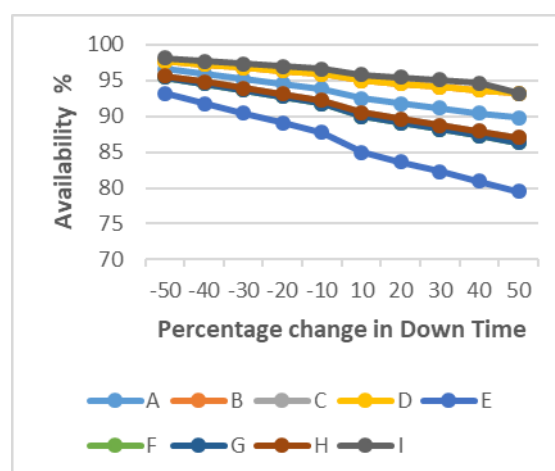
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|    | A               | B             | C           | D          | E            | F           | G       | H      |
|----|-----------------|---------------|-------------|------------|--------------|-------------|---------|--------|
| 1  | EQUIPMENT       | DOWNTIME(min) | TOTAL COUNT | GOOD COUNT | AVAILABILITY | PERFORMANCE | QUALITY | OE (%) |
| 2  | PADDY PARBOILER | 45            | 5000        | 4850       | 0.931818182  | 0.162601626 | 0.97    | 14.69  |
| 3  | PADDY DRYER     | 30            | 5000        | 4850       | 0.954545455  | 0.158730159 | 0.97    | 14.69  |
| 4  | RICE CLEANER    | 57            | 5000        | 4850       | 0.913636364  | 0.165837479 | 0.97    | 14.69  |
| 5  | PADDY DESTONER  | 30            | 5000        | 4850       | 0.954545455  | 0.158730159 | 0.97    | 14.69  |
| 6  | DEHUSKER        | 90            | 5000        | 4850       | 0.863636364  | 0.175438596 | 0.97    | 14.69  |
| 7  | RICE WHITER     | 60            | 5000        | 4850       | 0.909090909  | 0.166666667 | 0.97    | 14.69  |
| 8  | RICE POLISHER   | 60            | 5000        | 4850       | 0.909090909  | 0.166666667 | 0.97    | 14.69  |
| 9  | COLOUR SORTER   | 57            | 5000        | 4850       | 0.913636364  | 0.165837479 | 0.97    | 14.69  |
| 10 | PACKAGING SCALE | 25            | 5000        | 4850       | 0.962121212  | 0.157480315 | 0.97    | 14.69  |
| 11 | AVERAGE (%)     |               |             |            | 92.35690236  | 16.42210163 | 97      | 14.69  |

From the data of the rice packaging company in Table 1, the downtime, break-time (60 minutes) and shift length Time (720 minutes) were varied from negative (-) 50% to positive (+) 50 % to observe the changes on the parameter processes. This was captured in the Figure 1, 2, 3, 4, 5, and 6.

In Figure 1, from the graph all the equipment gives different results on availability because the availability changes when the downtime varies from

one percentage to another in each of the equipment. As the varying downtime at 50 % will give different availability on all the equipment and varying at 40 % will give different availability on each of the equipment. As the downtime percentage increases in the x axis the availability percentage decreases in y axis on each equipment. The varying downtime have relationship with availability which will affect the productivity of the rice packaging company. The legend lines represent the varying availability of the equipment A, B, C, and as shown in Figure 1.

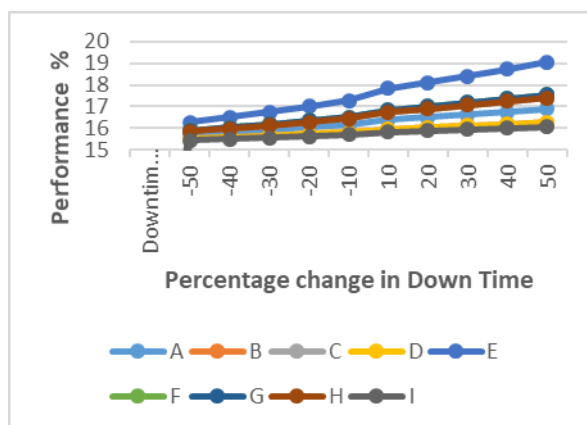


**Fig. 1:** *Effect of Changes in Downtime on Component Availability.*

### Varying Downtime for Process Parameter on Performance

In Figure 2, the graph represents the varying percentage of the downtime and the performance. From the variation on each equipment, the downtime and performance have correlation from the

equipment A to I which is negligible and the effect of this slight increase is due equipment upgrade and repair. The legend lines represent the varying performance of equipment A, B, C, ... and I as shown in Figure 2.

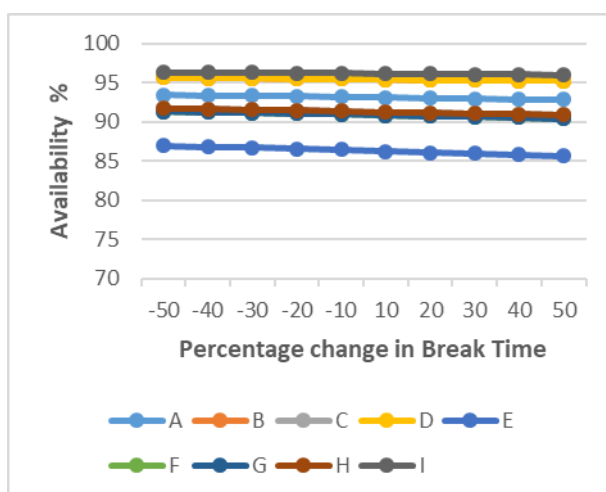


**Fig. 2:** Effect of Changes in Downtime on Component Performance.

### Varying Break-Time for Process Parameter on Availability

In Figure 3 below, the graph represents the varying percentage of the Break-time and the availability. From the variation on each equipment, the break-time and availability

have correlation from the equipment A to I, as the breaktime increases the availability decreases. The legend lines represent the varying availability of the equipment A, B, C, ... and I as shown in Figure 3.



**Fig. 3:** Effect of Changes in Break Time on Component Availability

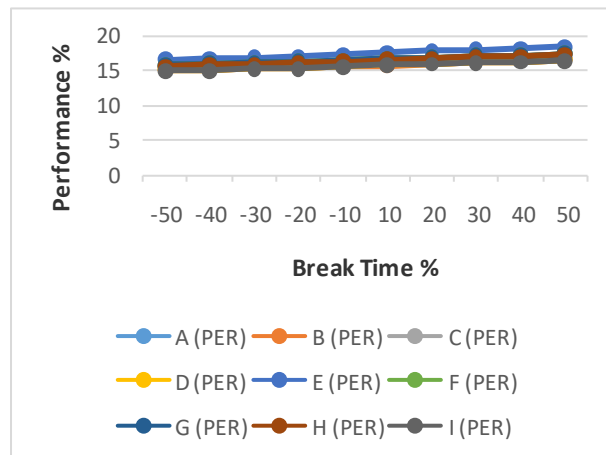
### Varying Break-Time for Process Parameter on Performance

In Figure 4 below, the graph represents the varying percentage of the Break-time and the performance. From the variation on

each equipment, the break-time and performance have correlation from the equipment A to I, as the break-time increases the performance slightly increase which the slight increase is due to

preventive maintenance on the equipment. The legend lines represent the varying

performance of equipment A, B, C, ... and I as shown in Figure 4.

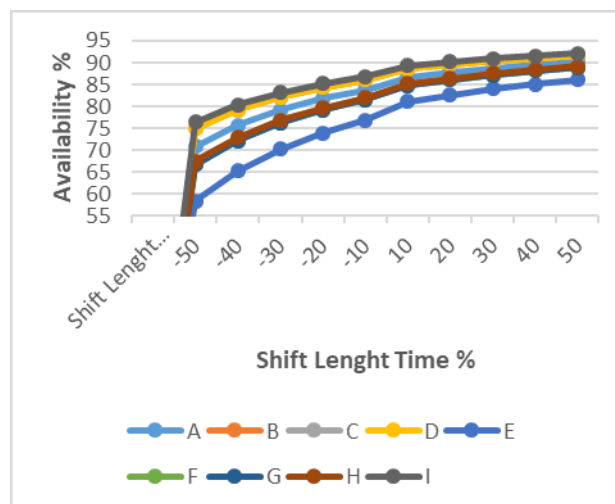


**Fig. 4:** Effect of Changes in Break Time on Component Performance.

### Varying Shift Length Time for Process Parameter on Availability

In Figure 5, the graph represents the correlation of the varying shift length time of each equipment and the availability. From the graph, each equipment result of shift length varies with availability as from

percentage to another. The varying shift length time has correlation with the availability, as the shift length time increases the availability increases. The legend line represents the availability of the equipment A, B, C, ... and I as shown in Figure 5.



**Fig. 5:** Effect of Changes in Shift Length Time on Component on Availability.

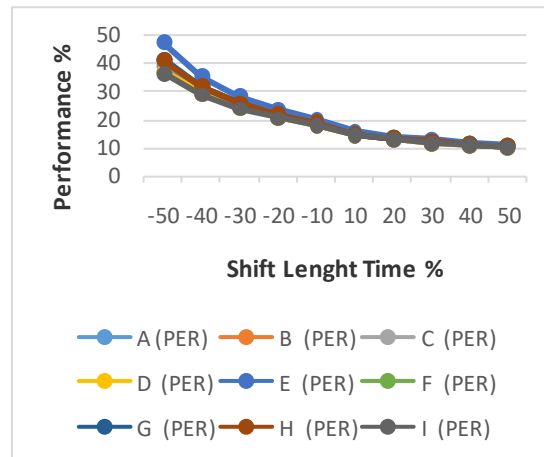
### Varying Shift Length Time for Process Parameter on Performance

In Figure 6, the graph represents the correlation of the varying Shift Length Time of each equipment and the performance. From the graph, all the equipment gives different results on performance because the performance

changes and shift length time varies from one percentage to another in each equipment. The varying shaft length time has correlation with performance, as shift length time increases the performance slightly decreases due to fatigue from long operational time of equipment which will affect the productivity of the rice

packaging company. The legend line represents the performance of the

equipment A, B, C, ... and I as shown in Figure 6.



**Fig. 6:** Effect of changes in Shift Length Time on component on performance.

### Results of Line Balancing Metric

In line balancing the list precedence of the rice packaging company was used to sketch the precedence diagram showing the tasks from the start to finish and the cycle time. Here from the precedence diagram the production line is not balanced. Adopting the Heuristic method, this helped to create work stations to show maximum takt time in each column to minimise idle time of equipment in each station. Task A and B were grouped in

work station 1, in work station 2, task C, D, and E were grouped together with cumulative time of 420 secs. In work station 3, task F, G, and H are grouped together with cumulative time of 420 secs. In the last work station 4, it has task I that has cumulative time of 120 seconds and idle time of 312 seconds. The whole tasks have been distributed to 4 work stations as the production line is balanced. The balanced production line is shown in Table 2.

**Table 2:** Table of Balanced Tasks on Work Stations.

| Assigned task | Cycle time (sec) | Cumulative time | Idle time | Work stations |
|---------------|------------------|-----------------|-----------|---------------|
| A             | 120              | 120             |           | 1             |
| B             | 240              | 360             | 70        |               |
| C             | 120              | 120             |           | 2             |
| D             | 120              | 240             |           |               |
| E             | 180              | 420             | 12        |               |
| F             | 180              | 180             |           | 3             |
| G             | 180              | 360             |           |               |
| H             | 60               | 420             | 12        |               |
| I             | 120              | 120             | 312       | 4             |

### Results of Line Efficiency Metric

The line efficiency was used to measure the improvement of the line balancing which is 76.39% using Equation 8, as this shows the time the equipment was active in assembly line.

Also, the delay balance time was calculated to be 23.61% using Equation 9, as this is the time equipment was not active in assembly line.



The line balancing was used to improve the production line of rice packaging company by 61.69%.

## CONCLUSION

### Conclusion for the 1<sup>st</sup> Objective

The evaluation data was used to get the process parameters Availability (92.36%), Performance (16.42%), and Quality (97%). This objective was achieved by calculating planned production time, run time and good count before evaluating the process parameters.

### Conclusion for the 2<sup>nd</sup> Objective

The evaluation of OEE is the total product of the process parameters from the 1st objective which is 14.7%.

This objective was achieved by evaluating the product of The components that determine the manufacturing line's overall equipment effectiveness are availability, performance, and quality.

### Conclusion for the 3<sup>rd</sup> Objective

The analysis of the key evaluation data like Downtime, Breaktime, and Shift Length Time shows that they have correlation with process parameters except Quality.

### Conclusion for the 4<sup>th</sup> Objective

The improvement of the production line was done using Line Balancing which has four workstations, Takt time of 432 secs, Total cycle time of 1320 secs which increased efficiency from 14.7% to 76.39%.

The objective was achieved using the Heuristic method to create workstations where the production tasks were assigned to it with equal takt time as the production line was improved by 61.69%.

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